# Practical 3

# Practice Lab Assignment

3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

**Input Format:**

- The first line contains two space-separated integers,  and , representing the number of rows and the number of columns.
- The next  lines contain  space-separated integers representing the elements of the array, entered row by row.

**Output Format:**

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

**Note:** Use reshape() function to reshape the input array with the specified number of rows and columns.

CODE:
```
import numpy as np
rows, cols =map(int,input().split())

elements=[]
for _ in range(rows) :
        elements.extend(map(int,input().split()))

array= np.array(elements).reshape(rows,cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```

| Average time | Maximum time |
|---|---|
| **0.025 s** | **0.044 s** |
| 25.25 ms | 44.00 ms |

2 out of 2 shown test case(s) passe

10 out of 10 hidden test case(s) pas

Test case 1 44 ms Debug

| Expected output | Actual output |
|---|---|
| 3 4 | 3 4 |
| 1 2 3 4 | 1 2 3 4 |
| 5 6 7 8 | 5 6 7 8 |
| 9 10 11 12 | 9 10 11 12 |
| [[·1··2··3··4] | [[·1··2··3··4] |
| ·[·5··6··7··8] | ·[·5··6··7··8] |
| ·[·9·10·11·12]] | ·[·9·10·11·12]] |
| 2 | 2 |
| (3,·4) | (3,·4) |
| 12 | 12 |

Test case 2 11 ms Debug

| Expected output | Actual output |
|---|---|
| 0 0 | 0 0 |
| [] | [] |
| 2 | 2 |
| (0,·0) | (0,·0) |
| 0 | 0 |

**Lab Assignment**

3.2.1. Numpy: Matrix Operations

The given code takes two  matrices, , and , as input from the user and converts them into NumPy arrays.

**Task:**

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

**Input Format:**

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

**Output Format:**

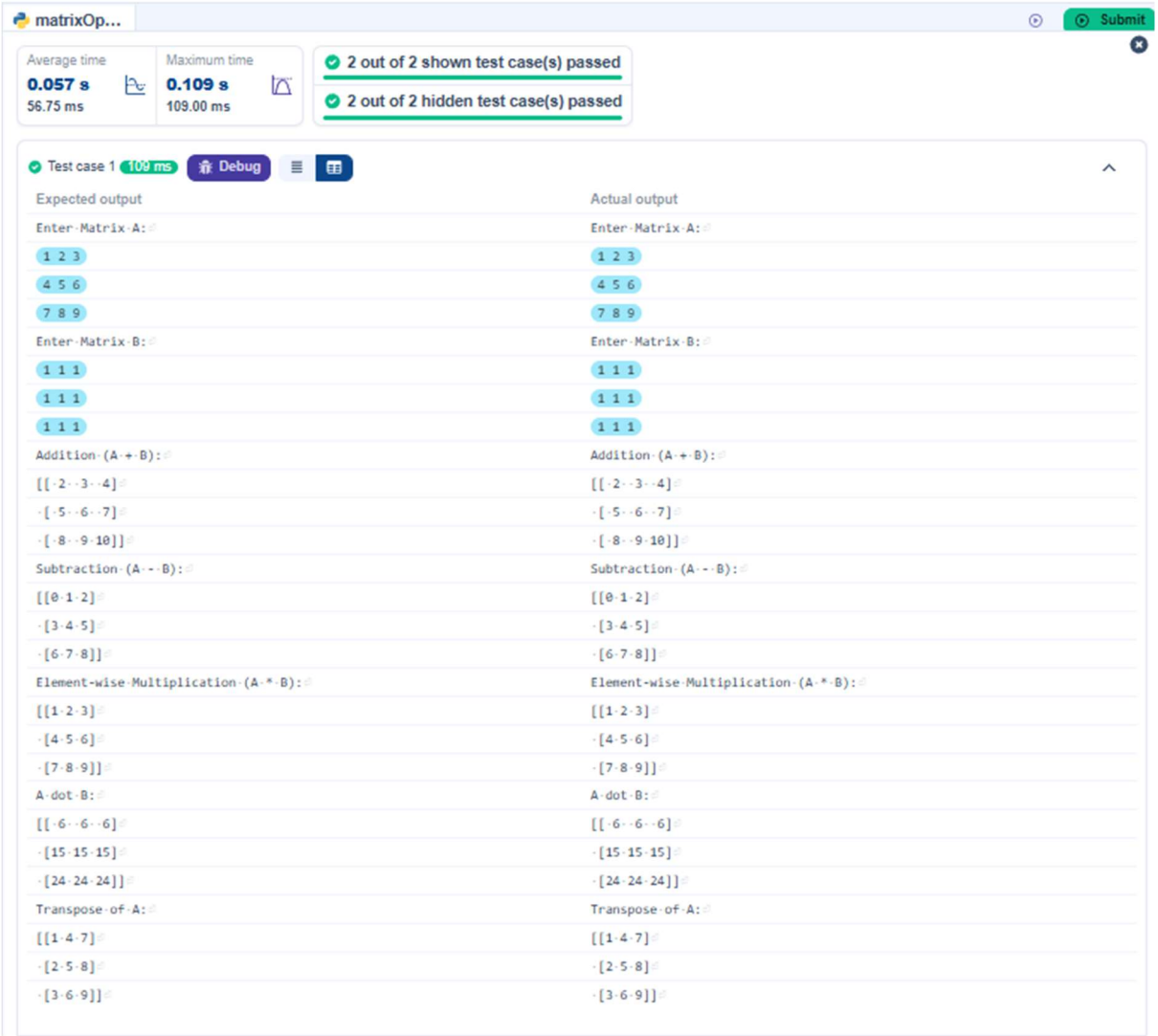
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

CODE:

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])



# Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)
# Subtraction
substraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(substraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)



# Matrix multiplication (dot product)

matrix_multiplication_result =np.dot(matrix_a, matrix_b)
print("A dot B:")
```

```
print(matrix_multiplication_result)

# Transpose

transpose_a = matrix_a.T

print("Transpose of A:")

print(transpose_a)
```



Test case 2 43 ms Debug

| Expected output | Actual output |
|---|---|
| Enter Matrix A: | Enter Matrix A: |
| 2 4 8 | 2 4 8 |
| 9 6 5 | 9 6 5 |
| 7 8 9 | 7 8 9 |
| Enter Matrix B: | Enter Matrix B: |
| 10 12 13 | 10 12 13 |
| 14 15 16 | 14 15 16 |
| 17 18 19 | 17 18 19 |
| Addition (A + B): | Addition (A + B): |
| [[12 16 21] | [[12 16 21] |
| [23 21 21] | [23 21 21] |
| [24 26 28]] | [24 26 28]] |
| Subtraction (A - B): | Subtraction (A - B): |
| [[ -8 -8 -5] | [[ -8 -8 -5] |
| [ -5 -9 -11] | [ -5 -9 -11] |
| [-10 -10 -10]] | [-10 -10 -10]] |
| Element-wise Multiplication (A * B): | Element-wise Multiplication (A * B): |
| [[ 20 48 104] | [[ 20 48 104] |
| [126 90 80] | [126 90 80] |
| [119 144 171]] | [119 144 171]] |
| A dot B: | A dot B: |
| [[212 228 242] | [[212 228 242] |
| [259 288 308] | [259 288 308] |
| [335 366 390]] | [335 366 390]] |
| Transpose of A: | Transpose of A: |
| [[2 9 7] | [[2 9 7] |
| [4 6 8] | [4 6 8] |
| [8 5 9]] | [8 5 9]] |

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

08:24

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

**Input Format:**

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

**Output Format:**

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
```

| Average time | Maximum time |
|---|---|
| 0.052 s | 0.066 s |
| 52.00 ms | 66.00 ms |

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 68 ms Debug

| Expected output | Actual output |
|---|---|
| Enter Array1: | Enter Array1: |
| 1 2 3 | 1 2 3 |
| 4 5 6 | 4 5 6 |
| 7 8 9 | 7 8 9 |
| Enter Array2: | Enter Array2: |
| 4 5 6 | 4 5 6 |
| 7 8 9 | 7 8 9 |
| 10 11 12 | 10 11 12 |
| Horizontal Stack: | Horizontal Stack: |
| [[ 1 2 3 4 5 6] | [[ 1 2 3 4 5 6] |
| [ 4 5 6 7 8 9] | [ 4 5 6 7 8 9] |
| [ 7 8 9 10 11 12]] | [ 7 8 9 10 11 12]] |
| Vertical Stack: | Vertical Stack: |
| [[ 1 2 3] | [[ 1 2 3] |
| [ 4 5 6] | [ 4 5 6] |
| [ 7 8 9] | [ 7 8 9] |
| [ 4 5 6] | [ 4 5 6] |
| [ 7 8 9] | [ 7 8 9] |
| [10 11 12]] | [10 11 12]] |

Test case 2 45 ms Debug

| Expected output | Actual output |
|---|---|
| Enter Array1: | Enter Array1: |
| -1 -2 -3 | -1 -2 -3 |
| -4 -5 -6 | -4 -5 -6 |
| -7 -8 -9 | -7 -8 -9 |
| Enter Array2: | Enter Array2: |
| 10 11 12 | 10 11 12 |
| 13 14 15 | 13 14 15 |
| 16 17 18 | 16 17 18 |
| Horizontal Stack: | Horizontal Stack: |
| [[-1 -2 -3 10 11 12] | [[-1 -2 -3 10 11 12] |
| [-4 -5 -6 13 14 15] | [-4 -5 -6 13 14 15] |
| [-7 -8 -9 16 17 18]] | [-7 -8 -9 16 17 18]] |
| Vertical Stack: | Vertical Stack: |
| [[-1 -2 -3] | [[-1 -2 -3] |
| [-4 -5 -6] | [-4 -5 -6] |
| [-7 -8 -9] | [-7 -8 -9] |
| [10 11 12] | [10 11 12] |
| [13 14 15] | [13 14 15] |
| [16 17 18]] | [16 17 18]] |

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

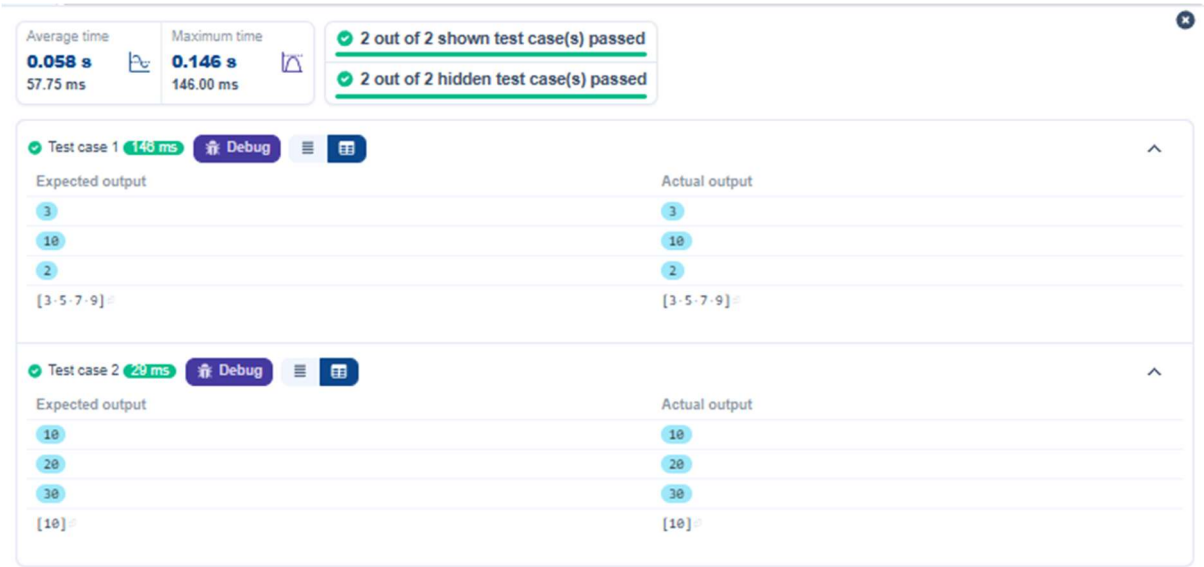
**Input Format:**

- The user will input three integer values: start, stop, and step, each on a new line.

**Output Format:**

- The program should print the generated sequence based on the input values.

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
n=np.arange(start,stop, step)

# Print the generated sequence
print(n)
```




3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

08:30

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

**1. Arithmetic Operations**:

- Compute the element-wise sum, difference, and product of the two arrays.

**2. Statistical Operations**:

- Calculate the mean, median, and standard deviation of array A.

**3. Bitwise Operations**:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i$ OR $B_i$).

**Input Format:**

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

**Output Format:**

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)
    # Arithmetic Operations
    sum_result = A + B
    diff_result = A -B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B
    or_result = A | B
    xor_result = A ^ B

  # Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
```

```
    print("Element-wise Product:", ' '.join(map(str, prod_result)))

    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

3.2.5. Numpy: Copying and Viewing Arrays

04:23

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

**Input Format:**

- A single line of space-separated integers.

**Output Format:**

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

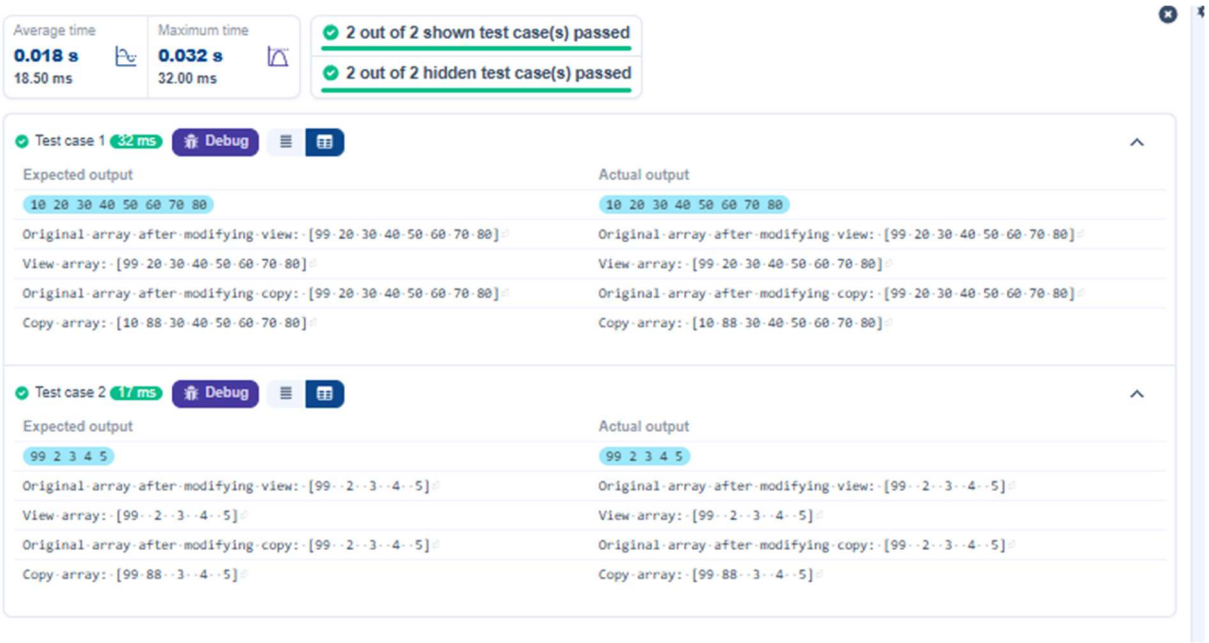
- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
```

```
view_array =original_array.view()

# Create a copy
copy_array =original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```



3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05:03

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

**1. Searching**: Find the indices where search_value appears in array1 and print these indices.

**2. Counting**: Count how many times count_value appears in array1 and print the count.

**3. Broadcasting**: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

**4. Sorting**: Sort array1 in ascending order and print the sorted array.

**Input Format:**

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

**Output Format:**

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

```
import numpy as np
```

```
# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)
# Broadcasting addition
c=array1 + broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
print(d)
```

| Average time | Maximum time |
|---|---|
| **0.035 s** | **0.061 s** |
| 34.75 ms | 61.00 ms |

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 61 ms Debug

| Expected output | Actual output |
|---|---|
| 1 1 1 2 2 2 | 1 1 1 2 2 2 |
| Value to search: 1 | Value to search: 1 |
| Value to count: 2 | Value to count: 2 |
| Value to add: 2 | Value to add: 2 |
| [0 1 2] | [0 1 2] |
| 3 | 3 |
| [3 3 3 4 4 4] | [3 3 3 4 4 4] |
| [1 1 1 2 2 2] | [1 1 1 2 2 2] |

Test case 2 27 ms Debug

| Expected output | Actual output |
|---|---|
| 1 2 3 4 5 | 1 2 3 4 5 |
| Value to search: 6 | Value to search: 6 |
| Value to count: 6 | Value to count: 6 |
| Value to add: 6 | Value to add: 6 |
| [] | [] |
| 0 | 0 |
| [ 7  8  9 10 11] | [ 7  8  9 10 11] |
| [1 2 3 4 5] | [1 2 3 4 5] |

3.2.7. Student Data Analysis and Operations

31:34

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.

- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

**Note:** Fill in the missing code to perform the above-mentioned operations.

CODE:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
print("All student Details:\n", a  )

# 2. print total students

print("Total Students:",len(a) )

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:,0]   )

# 4. Print subject 1 marks
```

```
print("Subject 1 Marks", a[:,1])

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2])      )

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:,3])    )

# 7. Print All subject marks
print("All subject marks:",  a[:,1:]     )

# 8. print Total marks of students
print("Total Marks", np.sum(a[:,1:], axis = 1)       )

# 9. print average marks of each student
print(np.round(np.mean(a[:,1:], axis = 1),1))

# 10. print average marks of each subject
print("Average Marks of each subject",  np.mean(a[:, 1:], axis = 0)     )

# 11. print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:, 1:3],axis=0 )        )

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3",  np.mean(a[:, [1, 3]], axis = 0)             )

# 13. print Roll number who got maximum marks in Subject 3
max_sub3_index = np.argmax(a[:, 3])
print("Roll no who got maximum marks in Subject 3", a[max_sub3_index , 0]          )
```

```
# 14. print Roll number who got minimum marks in Subject 2
min_sub2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_sub2_index, 0]     )

# 15. print Roll number who got 24 marks in Subject 2

print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2] == 24][:,0]}]"                )

# 16. print count of students who got marks in Subject 1 < 40
count_sub1_lt_40 = np.sum(a[:,1]<40)
print("Count of students who got marks in Subject 1 < 40",  count_sub1_lt_40            )

# 17. print count of students who got marks in Subject 2 > 90
count__sub2_gt_90 = np.sum(a[:,2]>90)
print("Count of students who got marks in Subject 2 > 90:",  count__sub2_gt_90           )

# 18. print count of students in each subject who got marks >= 90
count_each_subject_90plus = np.sum(a[:,1:]>=90 , axis = 0)
print("Count of students in each subject who got marks >= 90:",   count_each_subject_90plus
)

# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", a[:,0].astype(float)      )
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:] >= 90 , axis = 1)
)

# 20. Print S1 marks in ascending order
print(np.sort(a[:,1]))
```

# 21. Print S1 marks >= 50 and <= 90

```
s1_filteredv = a[(a[:,1]>=50) & (a[:,1]<=90)]
print(s1_filteredv)
print(a)
```

# 22. Print the index position of marks 79

```
indices_79 = np.where(a[:,1:]==79)[0]
print((indices_79,))
```

Average time
**0.103 s**
103.00 ms

Maximum time
**0.103 s**
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

| Expected output | Actual output |
|---|---|
| All student Details: | All student Details: |
| [[301. 67. 77. 88.] | [[301. 67. 77. 88.] |
| [302. 78. 88. 77.] | [302. 78. 88. 77.] |
| [303. 45. 56. 89.] | [303. 45. 56. 89.] |
| [304. 88. 98. 45.] | [304. 88. 98. 45.] |
| [305. 78. 88. 99.] | [305. 78. 88. 99.] |
| [306. 68. 78. 36.] | [306. 68. 78. 36.] |
| [307. 79. 12. 45.] | [307. 79. 12. 45.] |
| [308. 46. 45. 77.] | [308. 46. 45. 77.] |
| [309. 89. 32. 88.] | [309. 89. 32. 88.] |
| [310. 79. 24. 33.]] | [310. 79. 24. 33.]] |
| Total Students: 10 | Total Students: 10 |
| All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] | All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] |
| Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.] | Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.] |
| Min marks in Subject 2 12.0 | Min marks in Subject 2 12.0 |
| Max marks in Subject 3 99.0 | Max marks in Subject 3 99.0 |
| All subject marks: [[67. 77. 88.] | All subject marks: [[67. 77. 88.] |
| [78. 88. 77.] | [78. 88. 77.] |
| [45. 56. 89.] | [45. 56. 89.] |
| [88. 98. 45.] | [88. 98. 45.] |
| [78. 88. 99.] | [78. 88. 99.] |
| [68. 78. 36.] | [68. 78. 36.] |
| [79. 12. 45.] | [79. 12. 45.] |
| [46. 45. 77.] | [46. 45. 77.] |
| [89. 32. 88.] | [89. 32. 88.] |